

Time Series Analysis

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TS Forecasting & Unsupervised Analysis

Classical Forecasting and Unsupervised Methods for Time-Series

- **Detecting anomalies** and structural changes
- **Understanding patterns**, trends, and dependencies in temporal data
- **Applications** in finance, economics, industry, healthcare, and climate science
- **ML methods** for forecasting

Introduction to Unsupervised Methods

Why Unsupervised Learning?

Many time-series lack labels.

We want to discover:

- Patterns
- Anomalies
- Structural changes
- Similar sequences

without supervision.

Challenges in Time-Series

Time-series data is often:

- Noisy
- High-dimensional
- Non-stationary
- Continuously evolving

These characteristics complicate analysis.

Unsupervised methods for time-series

Time-series data differs from other data types because **observations depend on time**. The **same pattern** may have **different meanings** at **different** points in **time**, and time-series are often noisy and high-dimensional.

To make analysis easier, techniques such as **dimensionality reduction**, **wavelet analysis**, **filtering**, and **Fourier decomposition** can be used to reduce noise and extract important patterns. These preprocessing methods are commonly used in **anomaly detection** and **change point detection (CPD)**.

A **change point** (or **tipping point**) is a moment when the **behavior** of a **system changes** significantly. The example discussed in the text is climate change modeling, where researchers identified potential tipping points around **2010** and the **early 2020s**, representing major shifts in greenhouse gas emission trends and their impact on future global warming scenarios.

Unsupervised methods for time-series

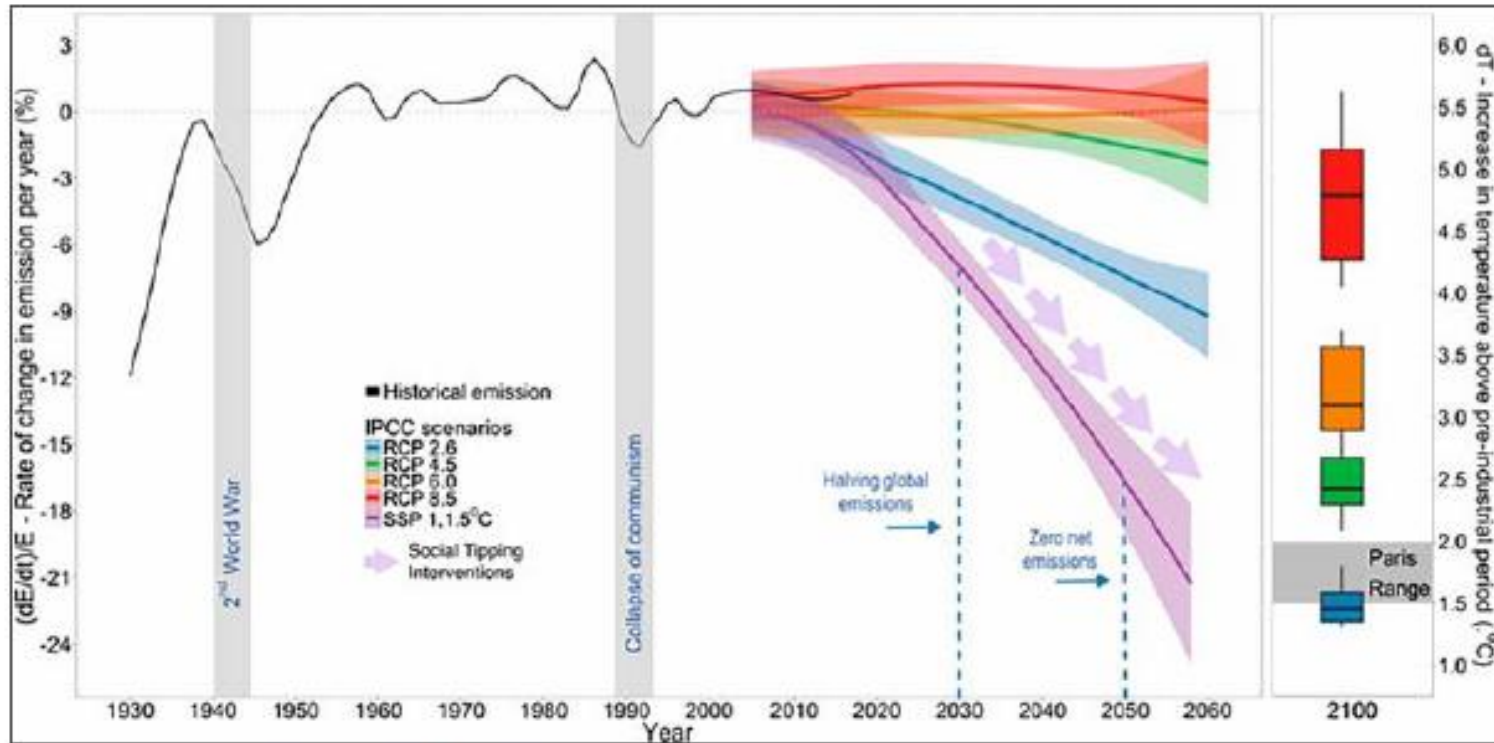


Figure 6.1: Possible change points based on greenhouse gas emissions

Starting around 2010–2020, several possible future paths are shown:

Scenario	Color	Interpretation
RCP 8.5	Red	Very high emissions ("business as usual")
RCP 6.0	Orange	Moderate-high emissions
RCP 4.5	Green	Moderate mitigation
RCP 2.6	Blue	Strong emission reductions
SSP 1.5°C	Purple	Very aggressive reductions to limit warming

The colored bands indicate uncertainty around each forecast.

Unsupervised methods for time-series

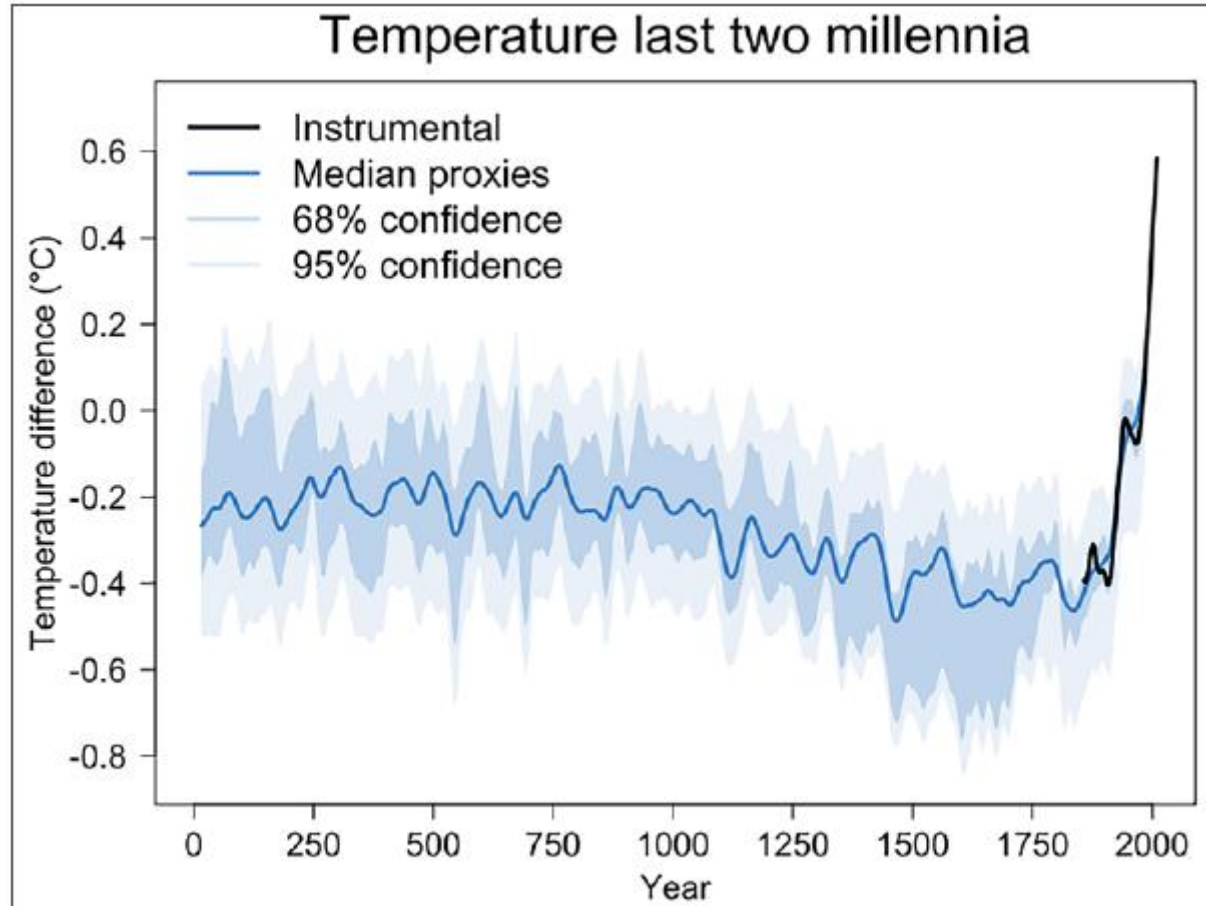


Figure 6.2: Change point in global temperatures: the beginning of the industrial age

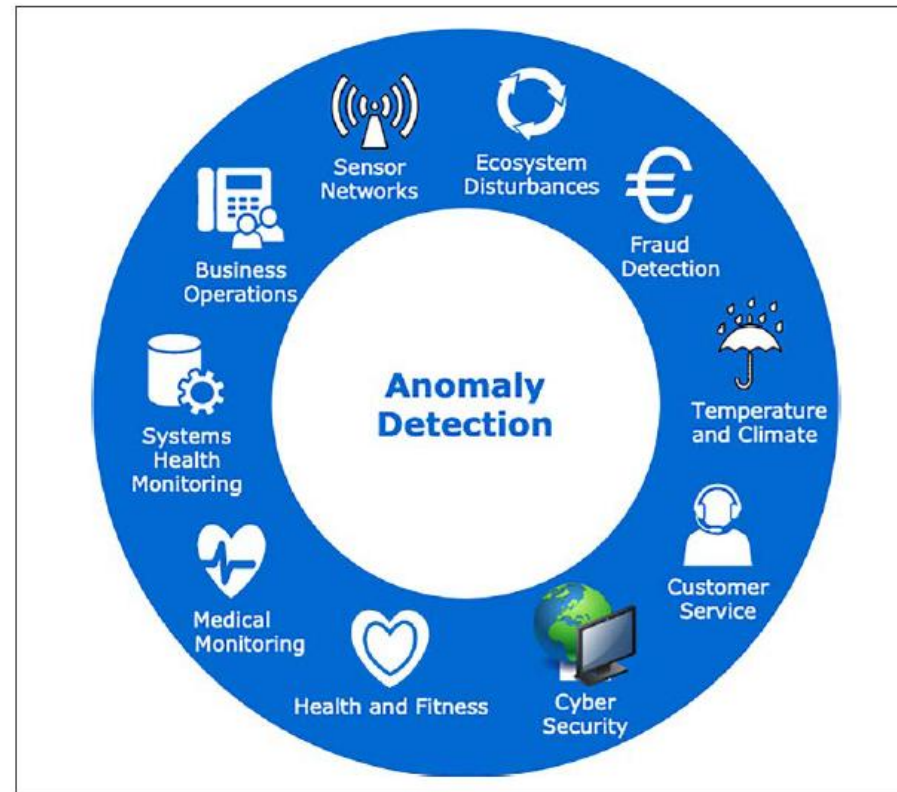
Humans can often identify anomalies and change points visually, especially with historical data. Automatic detection is more challenging and requires algorithms that balance **high detection accuracy** with **low false-positive rates**.

Anomaly Detection

Goal: Identify observations that significantly differ from normal behavior.

Applications:

- Fraud detection
- Equipment monitoring
- Cybersecurity
- Healthcare



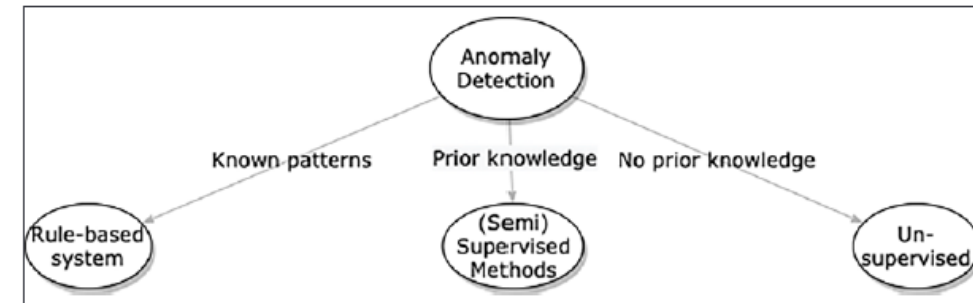
Anomaly Detection

Definition: An outlier is a data point that deviates so significantly from other observations that it could have been generated by a different mechanism.

Anomaly detection methods can be distinguished between univariate and multivariate methods.

Parametric anomaly detection methods, by the choice of their **distribution parameters** (for example, the arithmetic **mean**), place an assumption on the **underlying distribution** – often the Gaussian distribution.

These methods flag outliers, points that deviate from the model's assumptions.



Anomaly Detection – Z score

In the simplest case, we can define an outlier as follows as the z-score of the observation x_i with respect to the distribution parameters:

$$outlier(\hat{\mu}, \hat{\sigma}) = \left\{ x: \frac{|x_i - \hat{x}|}{\hat{\sigma}} > \epsilon \right\}$$

In this formula, $\hat{\mu}$ and $\hat{\sigma}$ are the estimated mean and standard deviation of the timeseries and x is the point that we want to test.

Finally, ϵ is a threshold dependent on the confidence interval that we are interested in – often, **2 or 1.96** are chosen for this, corresponding to a confidence interval of 95%.

In this way, **outliers** are points that **occur 5% or less of the time**.

The Hampel filter

The Hampel filter (also: Hampel identifier) is a special case for this, where the median and the median absolute deviation (MAD) are employed:

$$HF(x) = \left\{ x: \frac{x - \text{median}(x)}{1.4826 \text{ MAD}(x)} > \epsilon \right\}$$

In this equation, the sample mean is replaced by the (sample) median and the standard deviation by the MAD, which is defined as:

$$\text{Median}(|x_i - \text{median}(x)|)$$

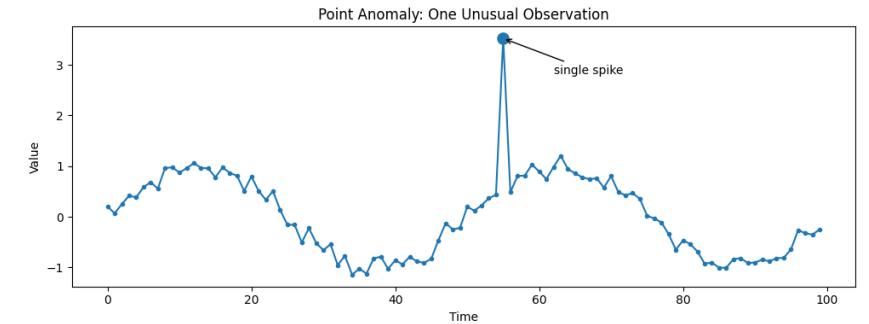
While these methods are restricted to **low-dimensional** or **univariate timeseries**, **distance-based** methods can handle much larger spaces. Distance-based anomaly detection methods effectively **cluster points into different groups**, where small groups will be labeled outliers.

→ In these methods, the choice of distance measure is crucial.

Types of Anomalies

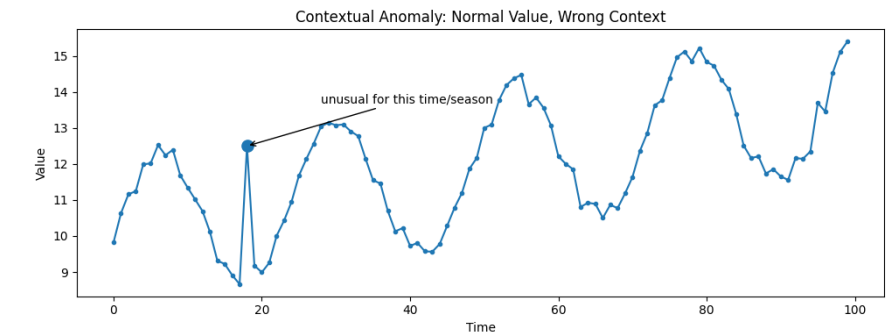
Point Anomalies

Single unusual observations.



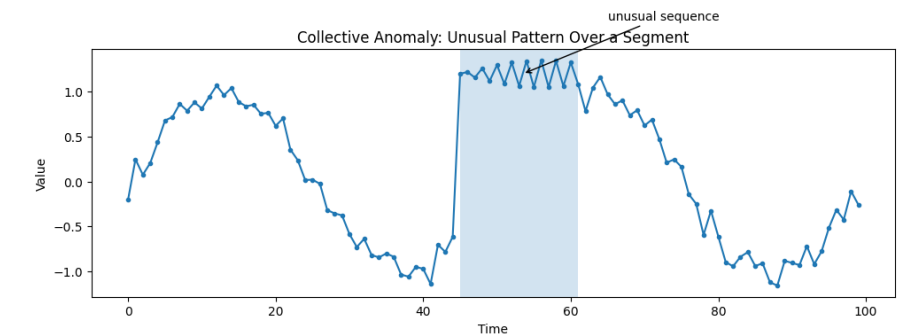
Contextual Anomalies

Unusual only in specific contexts.



Collective Anomalies

Groups of observations form unusual patterns.



Implementations

Library	Implementations	Maintainer	Stars
Prophet	Uncertainty interval around the estimated trend component from the forecast	Facebook Core Research	12.7k
PyOD	30 detection algorithms for multivariate time-series – from classical LOF (SIGMOD 2000) to COPOD (ICDM 2020)	Yue Zhao and others	4.5k
alibi-detect	Many anomaly detection algorithms – specific to time-series there are Likelihood Ratios, Prophet, Spectral Residual, Seq2Seq, Model distillation	Seldon Technologies Ltd	683
Scikit-Lego	Reconstruction through PCA/UMAP	Vincent D. Warmerdam and others	499
Luminaire	Luminaire Window Density Model	Zillow	371
Donut	Variational Auto-Encoder for Seasonal KPIs	Tsinghua Netman Lab	327
rrcf	Robust Random Cut Forest algorithm for anomaly detection on streams	Real-time water systems lab, University of Michigan	302
banpei	Hotelling's theory	Hirofumi Tsuruta	245
STUMPY	Matrix Profile algorithms for uni- and multivariate time-series such as STUMP, FLUSS, and FLOSS (also compare matrixprofile-ts)	TD Ameritrade	169
PySAD	More than a dozen algorithms for streaming outlier anomaly detection	Selim Yilmaz, Selim and Suleyman Kozat	98

Change Point Detection (CPD)

Definition:

Detect moments where the statistical properties of a series change.

A common problem with time-series is changes in the behavior of the observed system. Generally speaking, a **change point** signals an abrupt and **significant transition between states** in the **process** generating the series.

For example, the **trend** can suddenly **change**, and a **change point** can **signal** where the trend of the series changes.

This is well known under the guise of technical chart pattern analysis in trading.

Change Point Detection - Example

Suppose you are exercising now and **monitoring** your **heart** rate with a digital device. You **run** for a quarter mile, **walk** for ten minutes, then **run** for another quarter mile.

The heart rates may resemble the time series. It shows a cluster of **high** rates, a cluster of **low** rates, then back to **high** rates. The **abrupt** changes in the time series indicate that the source object has major **changes** in activity.

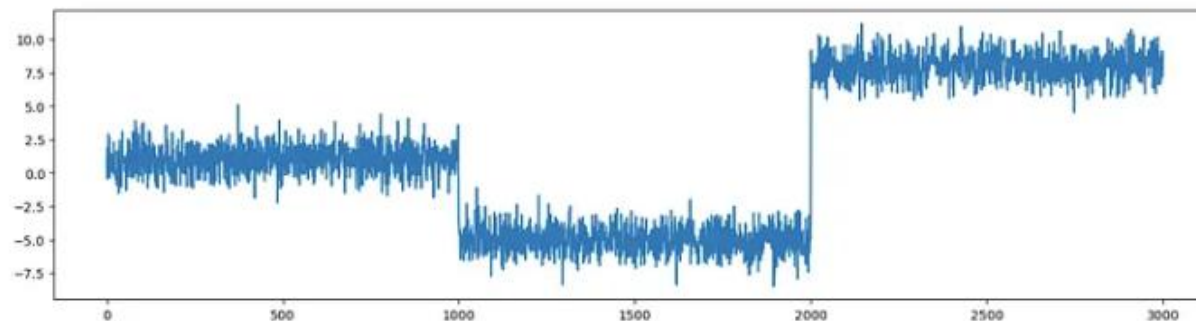


Figure (1)

Applications for CPD

- **Speech** recognition: Detection of **word** and **sentence boundaries**
- **Image** analysis: **Surveillance** on video footage of closed-circuit television
- **Fitness**: **Segmenting human activities** over time based on data from motion sensors from smart devices such as watches or phones
- **Finance**: Identifying **changes** to **trend patterns** that could indicate changes from bear to bull markets, or the other way around

CPD algorithms

CPD algorithms are composed of **three** components: **cost** functions, **search** methods, and **constraints**. We'll go through these in turn. Cost functions are distance functions that can be applied to a subsection of the time-series (multivariate or univariate).

An example for a cost function is **least absolute deviation (LAD)**, which is an estimator of a shift in the central point (mean, median, and mode) of a distribution defined as follows:

$$c(x_I) = \sum_{t \in I} |x_t - \hat{x}|$$

In this definition l is an index to a subsection in the time-series x , and $\hat{x}$ is the central point of x .

CPD algorithms

The search function then iterates over the time-series to detect change points. This can be done approximately, such as in window-based detection, bottom-up methods, or binary segmentation, or it can be exhaustive as in the case of dynamic programming or **Pruned Exact Linear Time (Pelt)**.

Pelt (Gachomo Dorcas Wambui and others, 2015) relies on pruning heuristics, and has a computational cost that is linear to the number of points of the time-series, $O(T)$. Dynamic programming methods have a much higher computational cost of $O(nT^2)$, where n is the maximum number of expected change points.

CPD algorithms

Finally, the constraint can come into play as a penalty in the search algorithm. This penalty term can encode a cost budget or knowledge of the number of change points that we would expect to find.

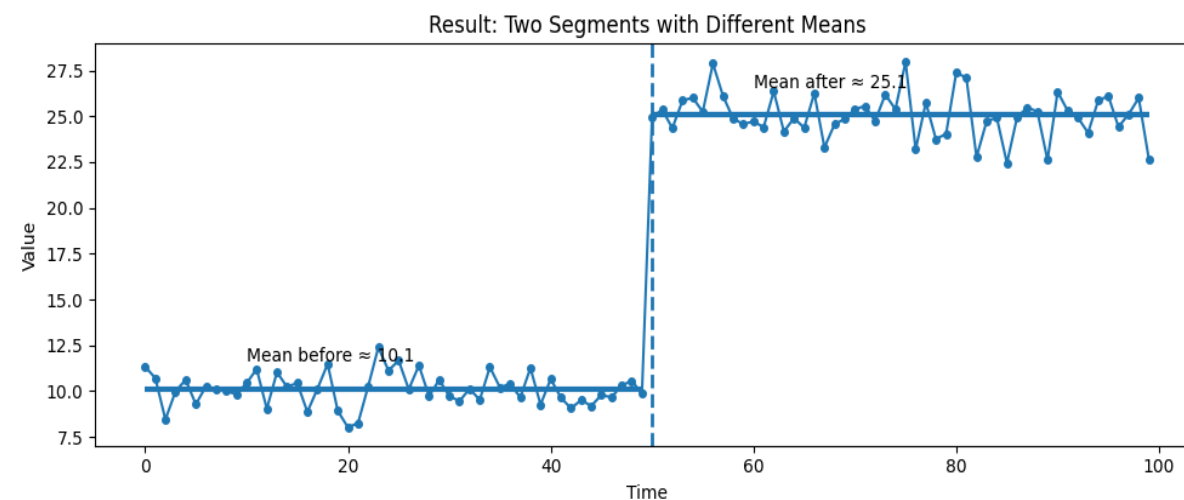
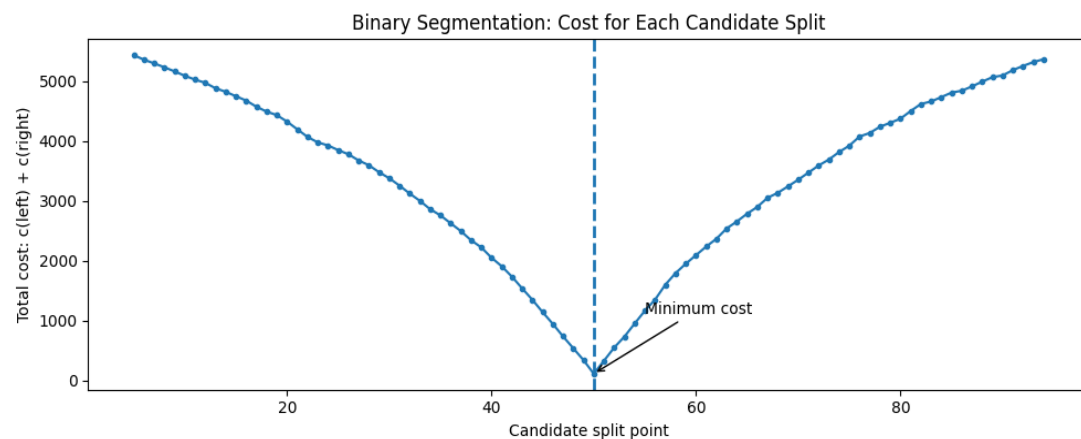
Binary segmentation ("*On Tests for Detecting Change in Mean*" by Ashish Sen and Muni S. Srivastava, 1975) falls into the category of window-based CPD. Binary segmentation is a greedy algorithm that minimizes the sum of costs the most as defined like this:

$$\hat{t} := \operatorname{argmin}_{1 \leq t < T-1} c(x_{0 \dots t}) + c(x_{t \dots T})$$

$\hat{t}$ is the found change point and $c()$ is a cost function similar to LAD, which we saw earlier in this section. The general idea is that when two subsequences are highly dissimilar, this indicates a change point.

Binary Segmentation

best split=the split that makes both parts internally consistent



Anomalies vs Change Points

CPD is related to segmentation, **edge detection**, **event detection**, and **anomaly detection**, and similar techniques can be applied to all these applications.

CPD can be viewed as very much like **anomaly** detection, since one way to identify **change** points is by **anomaly** scores from an anomaly detection algorithm.

From this perspective, **change points** are identical to highly **anomalous points**, and anything above a certain threshold corresponds to a change.

Real-World Examples

Applications include:

- Climate monitoring
- Industrial equipment
- Network traffic
- Financial markets
- Healthcare systems

Clustering Time-Series

Cluster analysis or clustering is the process of finding **meaningful groups** (clusters) of points or objects in a dataset based on their **similarity**. As the result of this unsupervised data mining technique, we want points in each cluster to be **similar** to each other, while being **different** to points in other clusters.

Clustering of time-series is **challenging** because each data **point** is a **period of time** (an ordered sequence). It has found application in diverse areas to **discover patterns** that empower time-series analysis, **extracting insights** from **complex datasets**.

Library	Implementations	Maintainer	Stars
tslearn	Time-Series K-Means, K-Shape clustering, KernelKMeans	Romain Tavenard	1.7k
river	DBStream, Time-Series K-Means, CluStream, DenStream, STREAMKMeans	Albert Bifet and others	1.7k

Distance Measures

Common similarity measures:

- Euclidean Distance
- Manhattan Distance
- Correlation Distance
- Dynamic Time Warping (DTW)

Choice of distance strongly affects clustering quality.

Euclidean Distance

Compares two time-series point by point:

$$d(x, y) = \sqrt{\sum_{t=1}^T (x_t - y_t)^2}$$

It works well when the series are **aligned in time**.

Example:

Series A: peak at time 20

Series B: peak at time 20

Example problem: two curves with the **same shape** but **shifted** slightly may look “far” under Euclidean distance.

Manhattan Distance

Similar to Euclidean, but uses absolute differences:

$$d(x, y) = \sum_{t=1}^T |x_t - y_t|$$

It is less sensitive to large single deviations than Euclidean distance.

Good for:

- Simple aligned time-series
- Cases where you want less emphasis on extreme errors

Bad for:

- Time shifts
- Different lengths

Correlation Distance

Measures whether two series move together, ignoring absolute scale:

$$d(x, y) = 1 - \text{corr}(x, y)$$

Two series can have **different magnitudes** but still be considered **similar** if their **patterns** rise and fall together.

Example:

Series A: [1, 2, 3, 4]

Series B: [10, 20, 30, 40]

Euclidean distance is large, but correlation distance is near zero because the shape is identical.

Dynamic Time Warping (DTW)

DTW calculates an optimal match by "**warping**" the **time** axis to find the **lowest** possible total **distance** between the two series.

For example, if one series reaches a peak slightly earlier than another, DTW can still match them as similar.

Conceptually:

Series A: peak at time 20

Series B: peak at time 25

Euclidean distance says: not very similar.

DTW says: similar shape, just shifted in time.

The standard DTW complexity is $O(T^2)$ for two sequences of length T .

Distance Measures

Which distance should you use?

Goal	Recommended distance
Same length, aligned series	Euclidean
Similar trend or co-movement	Correlation distance
Same shape but shifted in time	DTW
DTW but faster / more realistic	Constrained DTW
Shape-based clustering	SBD / K-Shape
Noisy or partially matching series	LCSS / EDR / ERP
Large datasets or interpretability	Feature-based distance

Evaluate Clustering Quality

Silhouette Score

Measures how similar a point is to its own cluster compared to other clusters.

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where:

- $a(i)$: average distance to points in the same cluster
- $b(i)$: average distance to points in the nearest other cluster

Range:

- +1 → very good clustering
- 0 → overlapping clusters
- 1 → wrong assignment

Important: for time-series, use the **same distance metric** used in clustering.
If you cluster with DTW, evaluate using DTW distance, not regular Euclidean distance.